

Nexa – Next Generation Office ERP System

A PROJECT REPORT

for

Mini Project (CA301P)

Session (2025-26)

Submitted by

Shantanu Yadav

202410116100193

Shivam Chaturvedi

202410116100198

Vineet Kumar

202410116100246

Shyam Sundar

202410116100207

**Submitted in partial fulfilment of the
Requirements for the Degree of**

MASTER OF COMPUTER APPLICATION

Under the Supervision of

**Ms. Shruti Aggarwal And Ms. Anita Yadav
Assistant Professor**



Submitted to

DEPARTMENT OF COMPUTER APPLICATIONS

KIET Group of Institutions, Ghaziabad

Uttar Pradesh-201206

DECEMBER 2025

DECLARATION

I hereby declare that the work presented in this report entitled “**Nexa - Next Generation Office ERP System**”, was carried out by me. I have not submitted the matter embodied in this report for the award of any other degree or diploma of any other University or Institute.

I have given due credit to the original authors/sources for all the words, ideas, diagrams, graphics, computer programs, experiments, results, that are not my original contribution. I have used quotation marks to identify verbatim sentences and given credit to the original authors/sources.

I affirm that no portion of my work is plagiarized, and the experiments and results reported in the report are not manipulated. In the event of a complaint of plagiarism and the manipulation of the experiments and results, I shall be fully responsible and answerable.

Shantanu Yadav

Shivam Chaturvedi

Vineet Kumar

Shyam Sundar

CERTIFICATE

Certified that **Shantanu Yadav 202410116100193, Shivam Chaturvedi 202410116100198, Shyam Sundar 202410116100207, and Vineet Kumar 202410116100246** has carried out the project work having “**NEXA- Next Generation Office ERP System**” (**Major-Project-CA301P**) for **Master of Computer Application** from Dr. A.P.J. Abdul Kalam Technical University (AKTU) (formerly UPTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other Institution.

**Ms. Shruti Aggarwal,
Ms. Anita Yadav
Assistant Professor
Department of Computer Applications
KIET Group of Institutions, Ghaziabad**

**Dr. Sachin Malhotra
Dean
Department of Computer Applications
KIET Group of Institutions, Ghaziabad**

NEXA - Next Generation Office ERP System

Shantanu Yadav
Shivam Chaturvedi
Vineet Kumar
Shyam Sundar

ABSTRACT

In the modern digital era, organizations must manage their internal operations efficiently to remain productive and competitive. However, many offices still depend on manual or partially computerized systems for handling employee records, attendance, and administrative activities. These traditional systems often result in issues such as data duplication, human errors, lack of transparency, delayed information access, and increased administrative workload. Such limitations negatively affect operational efficiency and decision-making. To overcome these challenges, the project titled “Office ERP System” has been developed as an integrated and centralized solution for effective office management.

The main objective of this project is to design and implement a centralized Enterprise Resource Planning system that automates key office operations, with a primary focus on employee management and attendance tracking. The system aims to reduce manual effort, improve data accuracy, enhance efficiency, and provide real-time access to reliable information. Role-based access control ensures data security by allowing administrators to manage records while enabling employees to securely view their personal and attendance details.

A structured development methodology was followed, starting with requirement analysis and system design using a modular architecture to ensure scalability and maintainability. Modern web technologies and a centralized database were used to develop a user-friendly system with proper authentication and validation mechanisms. After implementation, the system was thoroughly tested and found to be reliable and efficient. The Office ERP System is suitable for small to medium-scale organizations and can be further enhanced with features such as payroll, leave management, reporting, and analytics in the future.

ACKNOWLEDGEMENTS

Success in life is never attained single-handedly. My deepest gratitude goes to my project supervisor, Ms. Shruti Aggarwal And Ms. Anita Yadav for his guidance, help, and encouragement throughout my project work. Their enlightening ideas, comments, and suggestions.

Words are not enough to express my gratitude to Dr. Sachin Malhotra, Dean and Head, Department of Computer Applications, for his insightful comments and administrative help on various occasions.

Fortunately, I have many understanding friends, who have helped me a lot on many critical conditions.

Finally, my sincere thanks go to my family members and all those who have directly and indirectly provided me with moral support and other kind of help. Without their support, completion of this work would not have been possible in time. They keep my life filled with enjoyment and happiness.

Shantanu Yadav

Shivam Chaturvedi

Vineet Kumar

Shyam Sundar

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CHAPTER 1

INTRODUCTION

1.1 Overview

The Office ERP System is a dynamic web-based application designed to automate and streamline daily office operations. Developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), the system provides a centralized platform for managing employee records, attendance, and administrative activities efficiently. It aims to replace traditional manual or partially computerized office management methods with a reliable and user-friendly digital solution.

For administrators, the Office ERP System simplifies office management by enabling them to add, update, and manage employee information and monitor attendance records in real time. The system allows administrators to maintain accurate records, control user access, and ensure smooth administrative workflows, thereby reducing errors and paperwork associated with manual processes.

Employees benefit from an intuitive and secure interface that allows them to access their personal details and view attendance information easily. The centralized system ensures transparency and quick access to information, improving communication between employees and management. Authentication and role-based access control ensure that data remains secure and accessible only to authorized users.

Overall, the Office ERP System is a comprehensive solution that enhances operational efficiency, improves data accuracy, and supports better decision-making. By automating routine office tasks, the system addresses the challenges of modern office management and contributes to increased productivity in today's digital workplace.

Overview of the Project

Aspect	Description
Project Name	Office ERP System
Purpose	To automate and manage office operations such as employee records and attendance in an efficient and centralized manner.
Technology Stack	MERN Stack (MongoDB, Express.js, React.js, Node.js)
Target Users	Office administrators, employees and management staff.

Table 1.1 Overview of the Project

1.2 Problem Statement

Traditional office management methods often lead to several challenges that negatively impact efficiency and productivity. These challenges arise due to manual record keeping, fragmented systems, and lack of automation. The key issues addressed by this project include:

Manual and Time-Consuming Processes: Maintaining employee records and attendance manually requires significant effort and is prone to human errors.

Data Inaccuracy and Redundancy: Repetitive data entry often results in inconsistent and duplicate records.

Limited Accessibility: Important employee and attendance information is not easily accessible in real time.

Lack of Centralized System: Data stored across multiple files or systems makes management and tracking difficult.

Poor Administrative Control: Absence of role-based access reduces security and proper system monitoring.

Inefficient Reporting: Generating reports manually is slow and affects decision-making.

1.3 Objective

The primary objectives of the Office ERP System are as follows:

1. To provide a centralized and user-friendly platform for managing office operations.
2. To automate employee record management and attendance tracking.
3. To reduce manual effort, paperwork, and administrative workload.
4. To ensure accurate and consistent storage of employee and attendance data.
5. To implement secure user authentication and role-based access control.
6. To allow administrators to monitor and manage office activities efficiently.
7. To enable employees to securely access their personal and attendance information.
8. To provide real-time updates of attendance and employee records.
9. To improve transparency and communication within the organization.
10. To ensure data security and protect sensitive organizational information.
11. To design a scalable system that supports future enhancements.
12. To enhance overall productivity and operational efficiency of the organization.

1.4 Scope

The scope of the Office ERP System includes the following functionalities:

1. **Employee Management:** Administrators can add, update, and manage employee details in a centralized system.
2. **User Registration:** The system supports secure user accounts with role-based access for administrators and employees.
3. **Attendance Management:** Employees' attendance can be recorded, tracked, and viewed efficiently.
4. **Centralized Data Storage:** All office-related data is stored securely in a single database.
5. **Access Control:** Role-based authentication ensures secure and authorized access to system features.
6. **Real-Time Data Access:** Users can access updated employee and attendance information instantly.
7. **Administrative Control:** Administrators can monitor office activities and manage system data effectively.
8. **System Scalability:** The system is designed to support future expansion and additional modules.

1.5 Features

The Office ERP System provides the following key features:

- I. **Secure User Authentication:** Provides a login system with secure authentication for administrators and employees.
- II. **Role-Based Access Control:** Different access levels for administrators and employees to ensure data security.
- III. **Employee Management:** Allows administrators to add, update, view, and manage employee records efficiently.
- IV. **Attendance Tracking:** Enables accurate recording and monitoring of employee attendance.
- V. **Centralized Database:** Stores all employee and attendance data in a single, secure database.
- VI. **Real-Time Data Access:** Provides instant access to updated employee and attendance information.
- VII. **User-Friendly Interface:** Simple and intuitive interface for ease of use.
- VIII. **Data Accuracy and Consistency:** Reduces duplication and errors through automated data handling.
- IX. **Administrative Dashboard:** Allows administrators to monitor and control office operations effectively.
- X. **Scalable Architecture:** Designed to support future enhancements and additional modules.
- XI. **Improved Efficiency:** Automates routine tasks, reducing paperwork and administrative effort.
- XII. **System Security:** Protects sensitive organizational data through proper validation and access control.

1.6 Hardware Requirements

To run the Event Booking and Management System efficiently, the following hardware requirements are recommended:

- Processor: Dual-core processor (Intel i3 or equivalent)
- Ram: Minimum 8 GB
- Storage: At least 256 GB SSD
- Network: Stable internet connection (broadband recommended)

These specifications ensure optimal performance during development and usage.

Component	Minimum Requirements	Recommended Requirements
Processor	4 cores, 2.0 GHz	8 cores, 3.0 GHZ
RAM	8 GB	16 GB or more
Storage	100 GB SSD for system disk 100 GB HDD for data disk	256 GB SSD for system disk 500 GB HDD or larger for data storage
Network Card	1 Gbps	10 Gbps
Operating System	Windows Server 2016 or higher Linux (Ubuntu, CentOS)	Windows Server 2019 or higher Latest stable version of Linux

Table 1.2 Hardware Requirements

1.6 Software Requirements

The software requirements for the Event Booking and Management System include:

- Operating System: Windows 10 or later / macOS / Linux
- Node.js: Version 14.x or later
- MongoDB: Version 4.x or later
- React.js: Version 17.x or later
- Express.js: Version 4.x or later

Additionally, development tools such as Visual Studio Code or any preferred code editor should be installed to facilitate coding.

Software Component	Version/Requirement
Operating System	Window Server 2016 or higher / Linux (Ubuntu/CentOS)
Database	MongoDB (latest stable version)
Web Server	Apache HTTP Server (if using Linux) IIS (Internet Information Services)
Programming Language	Node.js (Latest stable version) React.js (Latest stable version) Express.js (Latest stable version)
Frameworks/Libraries	Bootstrap for responsive design jQuery for DOM manipulation
Security Software	SSL Certificate for secure transactions

Table 1.3 Software Requirements

CHAPTER 2

FEASIBILITY STUDY

2.1 Market Feasibility Study

A market feasibility study evaluates the demand for the Office ERP System within the target market. This study focuses on identifying whether organizations require an automated solution to manage employee records and attendance efficiently. With the increasing adoption of digital systems, offices are actively seeking centralized platforms that reduce manual work and improve productivity.

Key components of this study include:

Target Audience Analysis: Identifying the primary users of the system, which include office administrators, employees, and management personnel.

Competitive Analysis: Analysing existing office management tools to understand their limitations, such as lack of customization, high cost, or limited features.

Market Trends: Studying current trends in enterprise automation and digital workforce management to assess future growth potential.

The analysis indicates a strong demand for automated office management systems, making the Office ERP System a viable solution in the current market.

2.2 Technical Feasibility Study

The technical feasibility study examines whether the required technology and technical resources are available to successfully develop and deploy the Office ERP System. This includes evaluating software tools, system infrastructure, and development capabilities.

Key aspects include:

Technology Requirements: The system is developed using the MERN stack, including MongoDB for database management, Express.js and Node.js for backend development, and React.js for frontend implementation.

Infrastructure Assessment: The application can operate on standard hardware and cloud-based or local servers without the need for specialized infrastructure.

Technical Expertise: The development team possesses sufficient knowledge of web technologies and database management to implement the system effectively.

The study confirms that the chosen technology stack is suitable and supports scalability, maintainability, and performance requirements.

2.3 Financial Feasibility Study

The financial feasibility study analyses the cost-effectiveness of developing and implementing the **Office ERP System**. It compares estimated expenses with the expected benefits to determine economic viability.

Key components include:

- **Cost Analysis:** Includes expenses related to development, system deployment, hosting, and ongoing maintenance.
- **Cost Savings:** Automation reduces administrative workload, paperwork, and time consumption, resulting in long-term cost savings.
- **Return on Investment (ROI):** Improved productivity and efficient resource management justify the initial investment.

The analysis shows that the financial benefits outweigh the costs, making the project economically feasible.

2.4 Operational Feasibility Study

Operational feasibility evaluates whether the organization can effectively implement and sustain the Office ERP System within existing operational frameworks.

Key factors include:

- **Staffing Requirements:** The system does not require additional staff, as it is designed to be user-friendly and easy to manage.
- **Process Integration:** The system integrates smoothly with existing office workflows without significant disruption.

- **User Training Needs:** Minimal training is required due to the intuitive interface and simple navigation.

This study confirms that the system can be adopted smoothly with minimal operational challenges.

2.5 Legal Feasibility Study

The legal feasibility study identifies potential legal and regulatory concerns related to the implementation of the Office ERP System.

Key considerations include:

- **Data Privacy Regulations:** The system follows data protection practices to ensure the confidentiality and security of employee information.
- **Software Licensing:** All development tools and technologies used are open-source or properly licensed.
- **Compliance Requirements:** The system complies with organizational policies and applicable IT regulations.

This study concludes that the Office ERP System can be implemented without legal complications when standard compliance measures are followed.

Types of Feasibility	Key Findings
Market Feasibility	Sufficient demand exists for an automated office management and attendance system.
Technical Feasibility	The MERN stack supports a scalable, secure, and maintainable system.
Financial Feasibility	Project costs are justified by improved efficient and reduced administrative effort.
Operational Feasibility	The system can be integrated with minimal changes to existing office operations.
Legal Feasibility	Compliance with data protection and software licensing requirement is achievable.

Table 2.1 Types of Feasibility

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

3.1 Functionalities

The functionalities of the **Office ERP System – Employee Management System** cover a wide range of operations that support human resource and administrative activities within an organization. The system is designed to simplify and automate employee-related processes for HR administrators, managers, and employees.

At its core, the system allows **HR administrators** to create, update, and manage employee profiles. This includes maintaining personal details, job roles, departments, contact information, and employment status. HR users can also deactivate or update employee records when required. The system ensures that all employee information is stored in a centralized and secure database for easy access and management.

For **attendance and leave management**, the system provides functionalities to record daily attendance either through manual input or integration with biometric IoT devices. Employees can submit leave requests through the system, and managers are notified to approve or reject them. Once approved, the system automatically updates the employee's leave balance and reflects the changes in attendance records. Monthly attendance and leave reports can also be generated for monitoring and payroll purposes.

The system also supports **payroll processing**, where employee salaries are calculated based on attendance, approved leaves, and predefined salary structures. The ERP automatically generates payslips that employees can view and download through their accounts. This ensures accuracy, transparency, and timely salary processing.

For **performance management**, managers can conduct employee evaluations using predefined performance criteria. The system stores appraisal records, enabling organizations

to track employee growth, identify top performers, and make informed decisions regarding promotions and rewards.

In addition, the ERP provides **reporting and analytics** features that allow HR administrators and management to generate customizable reports related to attendance, leave, payroll, and employee performance. These reports help in analyzing workforce productivity and supporting strategic decision-making.

3.2 User and Characteristics

The **Office ERP System – Employee Management System** is designed to serve four primary user groups: **HR Administrators, Employees, Managers, and System Administrators**. Each user group has distinct roles, responsibilities, and system requirements that influence how the system is designed and operated.

- **HR Administrators** are responsible for managing employee-related operations within the organization. They require powerful tools to create, update, and maintain employee records, manage attendance, approve leave requests, generate payroll, and produce HR reports. Their characteristics include a need for accuracy, efficiency, and access to comprehensive employee data to ensure smooth HR operations.
- **Employees** are the primary users of the system who interact with it for personal and work-related information. They use the system to view and update their personal details, check attendance records, apply for leave, and access their payslips. Their characteristics include a preference for a simple and intuitive interface, quick access to their information, and transparency in leave and salary details.
- **Managers** play a supervisory role in the system. They are responsible for approving or rejecting employee leave requests and conducting performance evaluations. They also need access to team-related data such as attendance and performance records to make informed decisions. Their characteristics include a need for clear dashboards, timely notifications, and easy access to employee performance information.
- **System Administrators** are responsible for maintaining the overall ERP system. They manage user accounts, assign roles, ensure data security, and monitor system performance. Their characteristics include a strong focus on security, system reliability, and smooth operation of the entire platform.

3.3 Features of Project

The **Office ERP System – Employee Management System** is equipped with several key features that enhance its functionality, efficiency, and usability for HR staff, managers, and employees.

- **Employee Profile Management:**
The system allows HR administrators to create, update, and manage detailed employee profiles, including personal information, job roles, departments, and contact details.
- **Attendance Tracking:**
The system records daily employee attendance either through manual entry or through integration with biometric IoT devices. It also generates monthly attendance summaries.
- **Leave Management System:**
Employees can apply for leave online, and managers can approve or reject requests. The system automatically updates leave balances after approval.
- **Payroll Processing:**
The ERP calculates employee salaries based on attendance, approved leave, and predefined salary structures. It also generates downloadable payslips for employees.
- **Performance Appraisal:**
Managers can evaluate employee performance using predefined criteria. The system stores appraisal records for future reference and organizational planning.
- **Reporting and Analytics:**
HR administrators and management can generate reports related to attendance, leave, payroll, and employee performance to support decision-making.
- **Role-Based Access Control:**
Different users such as HR, employees, managers, and system administrators have access to features based on their roles, ensuring data security and proper authorization.
- **Notification System:**
Employees and managers receive email or system notifications for leave approvals, payroll generation, and important updates.

3.4 Features of Admin

The administrative features of the **Office ERP System – Employee Management System** are essential for maintaining control, security, and smooth operation of the entire platform. These features enable administrators to efficiently manage users, data, and system activities.

- **User Management:**
Administrators can create, update, deactivate, or delete user accounts for HR staff, employees, and managers. They can also assign roles and control access to different system features.
- **Employee Data Oversight:**
Admins have full access to all employee records and can monitor changes made by HR administrators, ensuring data accuracy and consistency across the system.
- **System Access Control:**
The system allows administrators to define permissions for different user roles such as HR, employees, and managers, ensuring that sensitive information is accessible only to authorized users.
- **Reporting and Monitoring:**
Administrators can view system-wide reports related to employee records, attendance, payroll, and performance. These reports help in tracking system usage and organizational data.
- **Security Management:**
Admins ensure data protection by enforcing security policies such as authentication, authorization, and secure data transmission. This helps in protecting sensitive employee and payroll information.
- **System Maintenance:**
The admin panel allows administrators to monitor system health, manage backups, and ensure that the ERP system runs smoothly without interruptions.
- **Notification and Communication:**
Administrators can send system-wide announcements, updates, or alerts to users regarding policy changes, system maintenance, or important organizational information.

3.5 Features of HR Administrative

HR administrators have access to specialized features designed to efficiently manage employee-related operations within the organization.

- **Employee Management Tools:**
HR administrators can add, edit, and deactivate employee profiles, including personal information, job roles, departments, and contact details.
- **Attendance Management:**
HR staff can monitor daily attendance records, verify attendance data from biometric IoT devices, and generate monthly attendance reports.
- **Leave Management Control:**
HR administrators can review employee leave requests, track leave balances, and ensure that leave policies are correctly applied across the organization.
- **Payroll Management:**
The system enables HR administrators to define salary structures, review payroll calculations, and generate payslips for employees.
- **Performance Monitoring:**
HR administrators can access employee performance appraisal records to support promotions, training, and organizational planning.
- **Employee Communication:**
HR staff can send notifications, announcements, or important updates to employees through the system.
- **Data and Report Generation:**
HR administrators can generate reports related to attendance, payroll, leave, and performance to support management decision-making.

3.6 Features of Employee

Employees benefit from a range of features designed to make it easy to access and manage their work-related information through the ERP system.

- **User-Friendly Interface:**
The system provides a simple and intuitive dashboard that allows employees to easily navigate through their profile, attendance, leave, and payroll information.
- **Personal Profile Management:**
Employees can view and update their personal details such as contact information and basic profile data.
- **Leave Application System:**
Employees can apply for leave online, check the status of their requests, and view their remaining leave balance.
- **Attendance Tracking:**
Employees can view their daily and monthly attendance records to ensure transparency and accuracy.
- **Payslip Access:**
The system allows employees to view and download their salary payslips anytime.
- **Real-Time Notifications:**
Employees receive alerts for leave approvals, payroll generation, and important system updates.
- **Performance Records:**
Employees can view their performance appraisal results and feedback provided by managers.

CHAPTER 4

SOFTWARE REQUIREMENT

4.1 Functional Requirement

Functional requirements define the specific behaviors and operations that the **Office ERP System – Employee Management System** must perform to support HR activities and employee management. These requirements ensure that the system meets the needs of HR administrators, managers, employees, and system administrators while delivering accurate and efficient services.

The key functional requirements of the system include:

- **User Authentication:**
The system must allow users to securely register, log in, and manage their accounts using role-based access. It should support password recovery and user verification to ensure system security.
- **Employee Management:**
HR administrators must be able to create, update, and deactivate employee records. This includes storing personal details, job roles, department information, and contact data.
- **Attendance and Leave Management:**
The system must record daily employee attendance and allow employees to apply for leave. Managers should be able to approve or reject leave requests, and the system must automatically update leave balances and attendance records.
- **Payroll Processing:**
The ERP system must calculate employee salaries based on attendance, approved

leaves, and predefined salary structures. It should also generate payslips that employees can access online.

- **Performance Management:**
Managers should be able to conduct performance evaluations, and the system must store appraisal records for future reference.
- **Admin Dashboard:**
Administrators must have access to a centralized dashboard to manage users, monitor system activity, generate reports, and ensure data security and system compliance.

These functional requirements ensure that the **Office ERP System – Employee Management System** provides a complete and reliable solution for managing employees and HR operations within an organization.

Requirement	Description
User Authentication	Users can securely register, log in, and manage their accounts with role-based access.
Employee Management	HR administrators can add, edit, and deactivate employee records.
Attendance & Leave Management	Employees can mark attendance and apply for leave; managers can approve or reject leave requests.
Payroll Processing	The system calculates salaries and generates payslips based on attendance and salary rules.
Performance Management	Managers can evaluate employee performance and store appraisal records.
Admin Dashboard	A centralized interface for managing users, monitoring system activity, and generating reports.

Table 4.1 Functional Requirements

4.2 Non-Functional Requirement

Non-functional requirements define the quality and performance standards of the **Office ERP System – Employee Management System**. These requirements describe how well the system should operate rather than what tasks it should perform. They ensure that the system is reliable, secure, scalable, and easy to use for all users.

- **Usability:**
The system should provide an intuitive and user-friendly interface so that HR staff, managers, and employees can easily use the system without extensive training. The design should be responsive and accessible across different devices.
- **Performance:**
The ERP system must be able to handle multiple users simultaneously without noticeable delays. Actions such as loading employee records, submitting leave requests, and generating payroll should be processed quickly and efficiently.
- **Scalability:**
The system architecture should support organizational growth. It should be capable of handling an increasing number of employees, records, and transactions without affecting system performance.
- **Security:**
All employee and payroll data must be protected using strong security mechanisms. Data transmission should be encrypted, and role-based access control must be enforced to prevent unauthorized access. The system should comply with data protection regulations.
- **Availability:**
The ERP system should be available to users at all times with minimal downtime. Scheduled maintenance should be planned and communicated to users in advance.
- **Backup and Recovery:**
The system should perform regular data backups to prevent data loss. In case of system failure, a recovery mechanism must be in place to restore data and system functionality quickly.

These non-functional requirements ensure that the **Office ERP System – Employee Management System** operates smoothly, securely, and reliably for all users.

Requirement	Description
Usability	Easy-to-use and responsive user interface for all users.
Performance	The system must handle multiple users and transactions without delays.
Scalability	The system should support growth in employees, data, and system usage.
Security	Strong data protection using encryption and role-based access control.
Availability	High system uptime with minimal downtime.
Backup and Recovery	Regular backups and fast recovery in case of system failure.

Table 4.2 Non-Functional Requirements

4.3 Design Goal

The design goals of the **Office ERP System – Employee Management System** guide the overall system architecture and user interface. These goals ensure that the system meets both functional and non-functional requirements while providing a smooth and efficient experience for all users.

Modularity:

The system should be designed in a modular manner so that different components such as employee management, attendance, payroll, and performance evaluation can be developed, updated, or maintained independently without affecting the entire system.

- **User-Centric Design:**

The interface should be simple, intuitive, and easy to use for HR staff, managers, and employees. User experience should be prioritized so that users can complete their tasks with minimal effort and confusion.

- **Integration Capabilities:**

The ERP system should support integration with external systems such as biometric

attendance devices, email and SMS services, and other enterprise applications to enable smooth data exchange and automation.

- **Responsive Design:**
The application must be fully responsive and accessible on desktops, laptops, tablets, and smartphones, ensuring users can access the system from any device.
- **Performance Optimization:**
The system should be designed to provide fast response times when loading employee data, processing payroll, or generating reports. Efficient coding practices and optimized database operations should be used.
- **Security and Data Protection:**
The design must ensure secure handling of sensitive employee and payroll data using encryption, authentication, and role-based access control.

By following these design goals, the **Office ERP System – Employee Management System** aims to deliver a secure, scalable, and user-friendly platform that effectively supports organizational HR and employee management needs.

4.4 Conclusion

In conclusion, the **Software Requirement Specification (SRS)** for the **Office ERP System Employee Management System** defines the essential functional, non-functional, and design requirements needed to build a reliable and efficient employee management platform. These requirements ensure that the system supports core HR operations such as employee management, attendance tracking, leave processing, payroll generation, and performance evaluation in a secure and organized manner.

ER Diagram

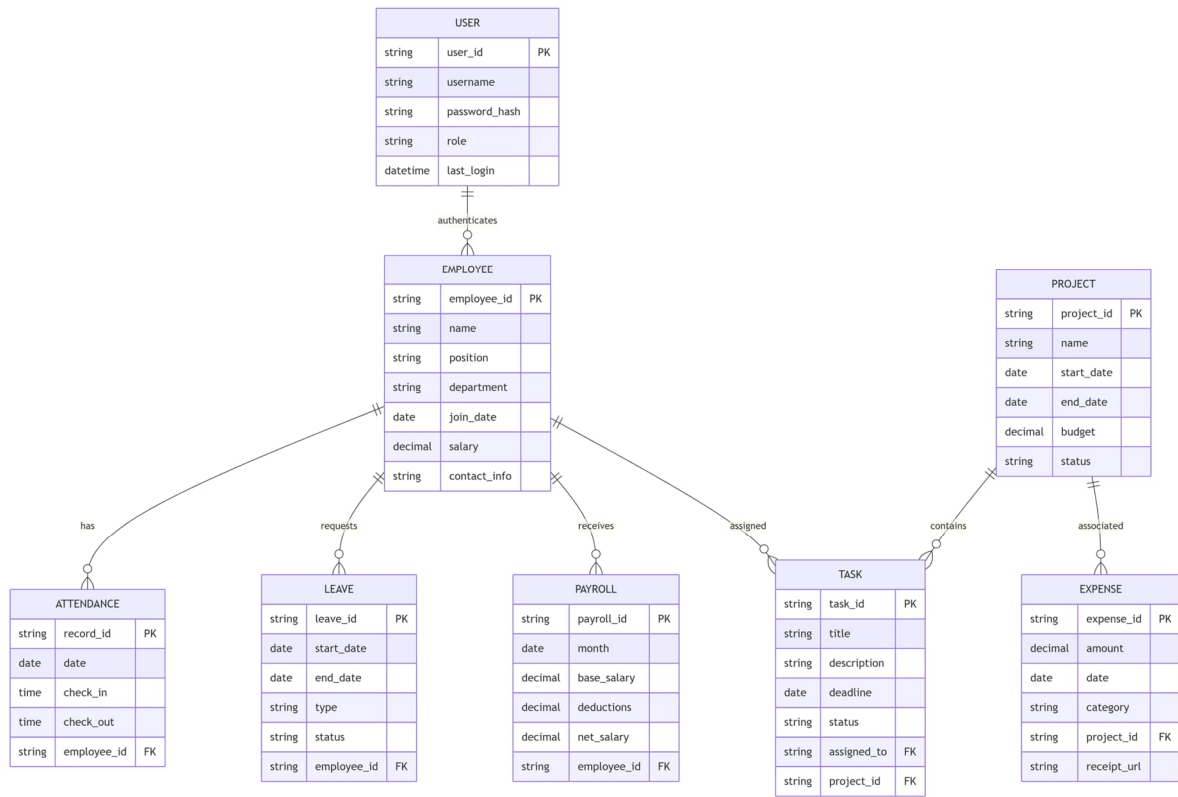


Fig 4.1 ER Diagram

Level 0 DFD

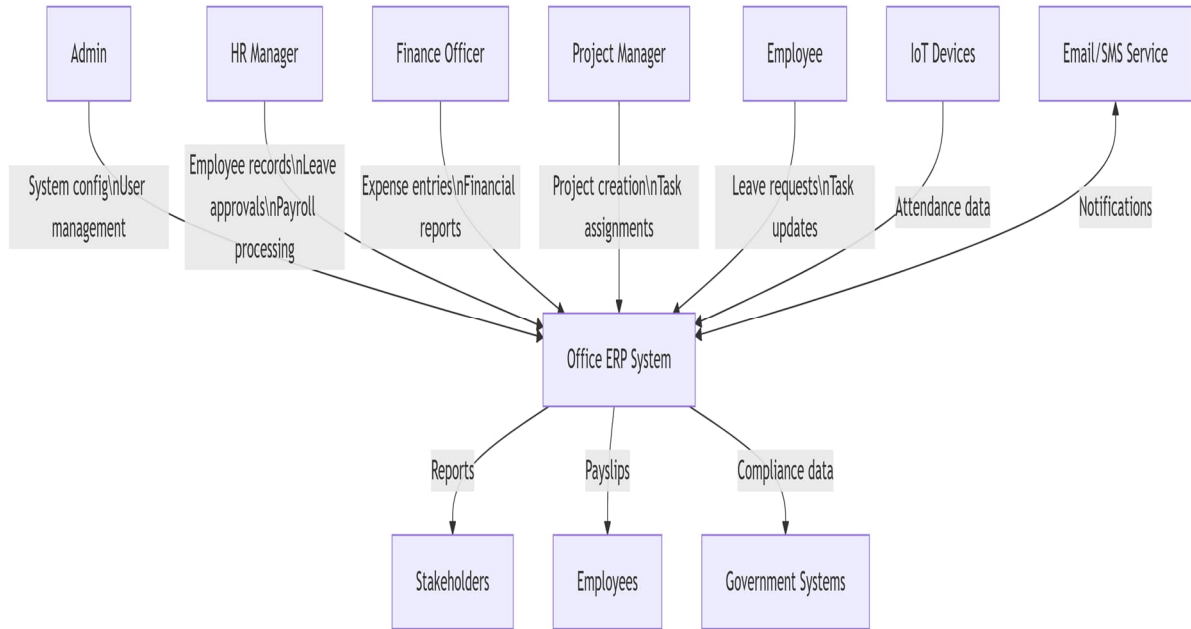


Fig 4.2 Level 0 Data Flow Diagram

Level 1 DFD

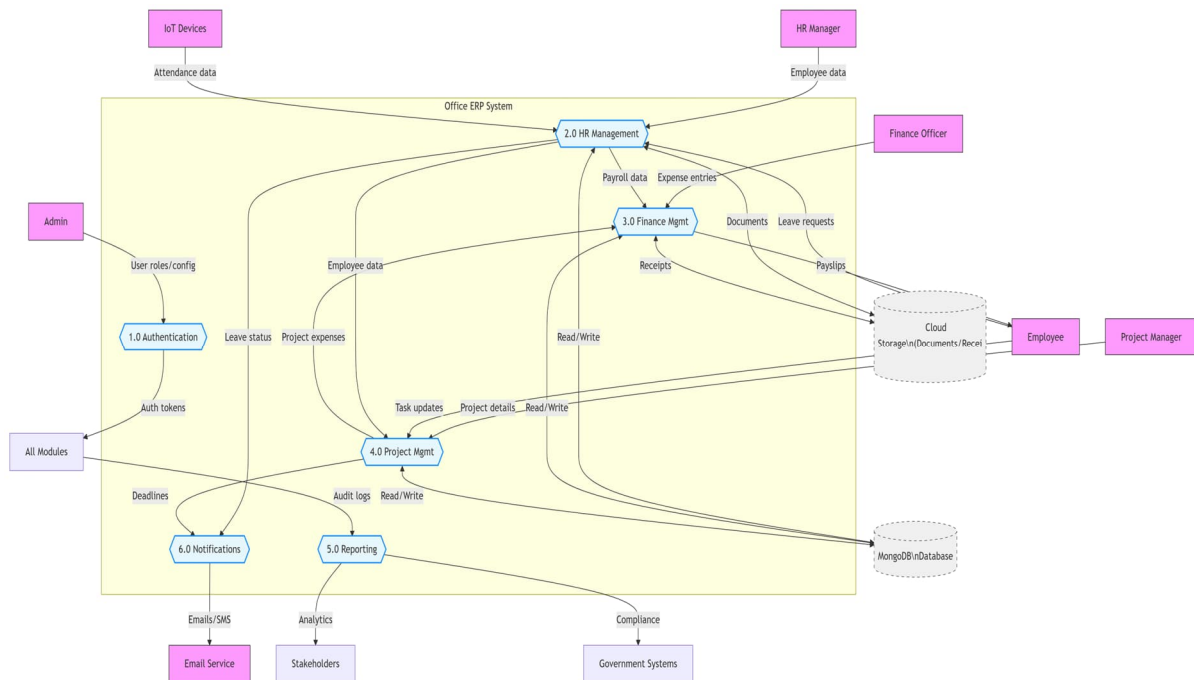


Fig 4.3 Level 1 Data Flow Diagram

CHAPTER 5

SYSTEM DESIGN

The design phase of the Software Development Life Cycle (SDLC) is a critical step in converting system requirements into a structured plan for implementation. For the **Office ERP System – Employee Management System**, this phase defines how the system will be built, how different modules will interact, and how data will flow through the system. The design ensures that the ERP is scalable, secure, and efficient while meeting all user and business requirements.

The system design is divided into three main parts: **Architectural Design, Interface Design, and Detailed Design**. Each of these plays an important role in ensuring that the ERP system is reliable and easy to use.

5.1 Architectural Design

Architectural design forms the backbone of the **Office ERP System – Employee Management System**. It defines the overall structure of the system and the way different modules interact with each other and with external services. The major components of the system include user authentication, employee management, attendance tracking, leave management, payroll processing, performance evaluation, and reporting.

The system follows a **client–server architecture**, where the frontend provides the user interface and the backend handles business logic and data processing. The frontend is developed using **React.js**, which provides a dynamic and responsive user experience. The backend is built using **Node.js and Express.js**, which handle API requests, authentication, and data processing. The system uses **MongoDB and MySQL** databases to store employee records, payroll data, and other organizational information.

Communication between the frontend and backend takes place through **RESTful APIs**. These APIs allow secure and structured data exchange for operations such as login, employee updates, leave requests, and payroll generation. **JWT-based authentication** is used to ensure secure access and role-based authorization.

The architectural design also supports integration with **biometric attendance devices** and external services such as email and SMS gateways for notifications. This allows the system to automatically capture attendance and send important alerts to employees and managers.

By using a modular architecture, each part of the ERP system (such as payroll, attendance, or performance management) can be developed, tested, and maintained independently. This improves system reliability, simplifies debugging, and allows future enhancements without affecting the entire application.

A well-defined architecture ensures that the **Office ERP System – Employee Management System** remains scalable, secure, and capable of supporting the growing needs of an organization.

5.2 Interface Design

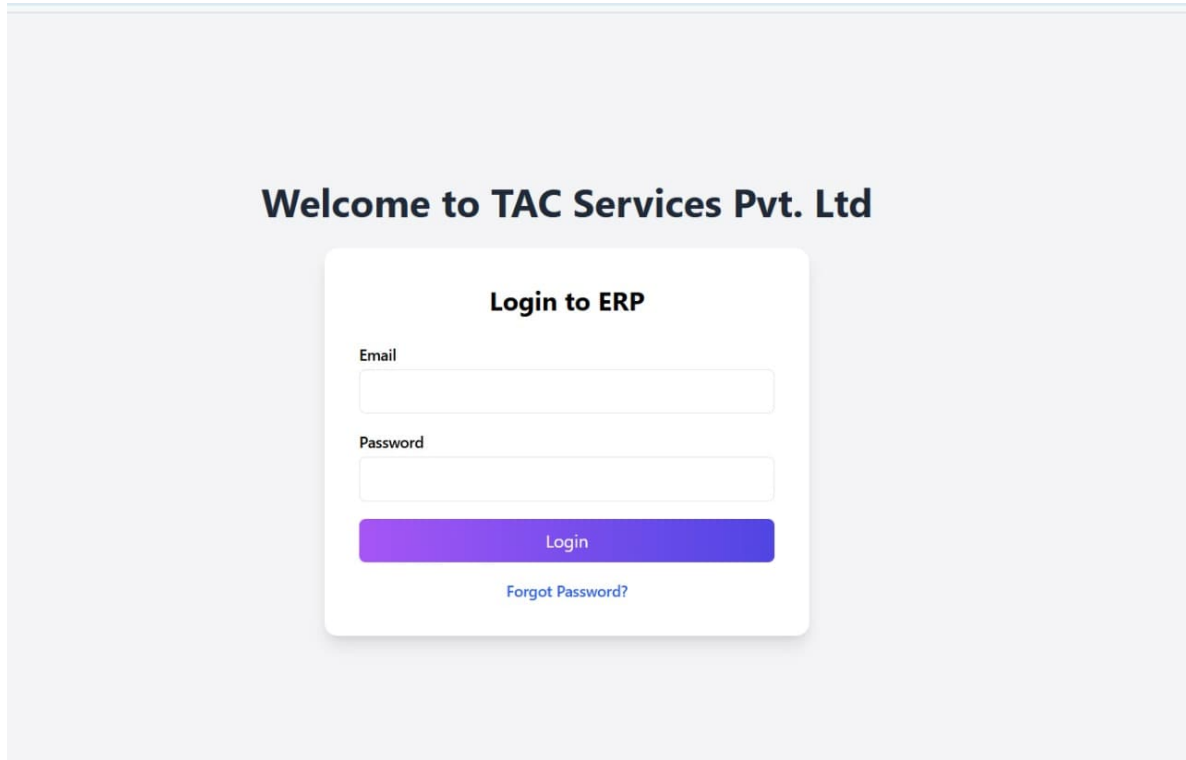
Interface design focuses on creating a simple, intuitive, and user-friendly experience for all users of the **Office ERP System – Employee Management System**, including HR administrators, managers, and employees. The main objective is to ensure that users can easily navigate the system and complete their tasks efficiently without confusion.

During this phase, screen layouts, wireframes, and prototypes are designed to visualize how different parts of the system will appear and function. For example, the employee dashboard should clearly display personal details, attendance records, leave status, and payslip information. Similarly, the HR and manager dashboards should provide quick access to employee management, leave approvals, payroll, and reports.

Usability testing plays an important role in interface design. Feedback from users such as HR staff and employees is used to improve the layout, navigation, and overall user experience before final deployment. This helps in reducing errors, improving productivity, and ensuring that the system is easy to use.

Accessibility is also considered in the interface design so that all users can interact with the system comfortably. Features such as clear fonts, simple navigation, and compatibility with different devices ensure that the ERP system can be used by a wide range of users.

By following these principles, the **Office ERP System – Employee Management System** provides a clean, responsive, and user-friendly interface that supports efficient employee and HR operations.



The image shows a web interface for logging into an ERP system. At the top, it says "Welcome to TAC Services Pvt. Ltd" in a bold, dark font. Below this is a white rectangular box with rounded corners. Inside the box, the title "Login to ERP" is centered. There are two input fields: one for "Email" and one for "Password". Below the password field is a blue button with the text "Login". At the bottom of the box, there is a link that says "Forgot Password?".

Fig 5.1 Sign in Page

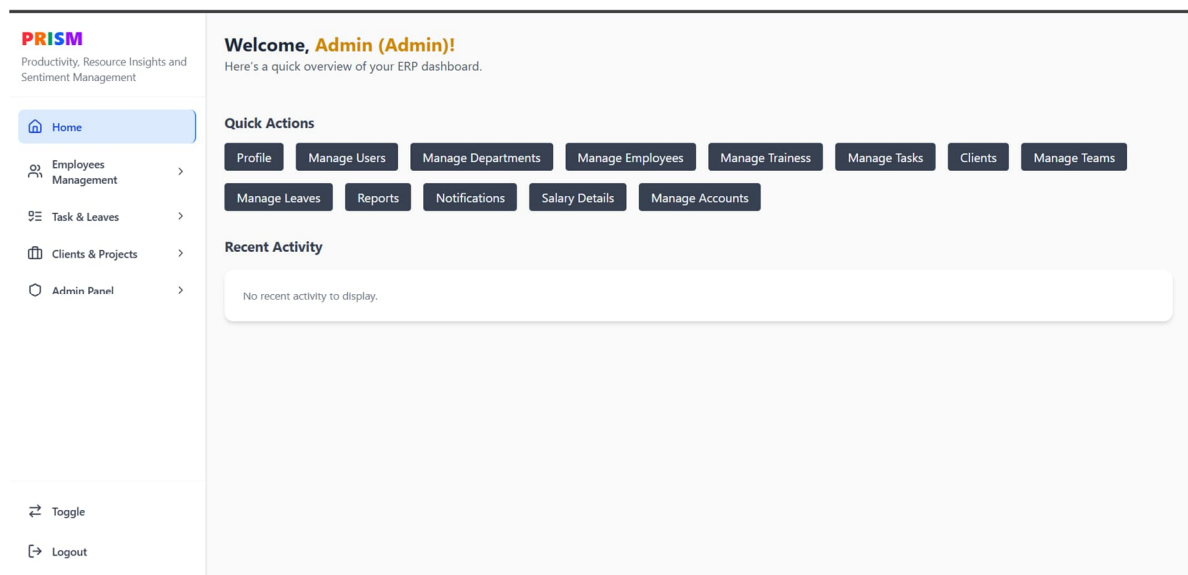


Fig. 5.2 Home Page

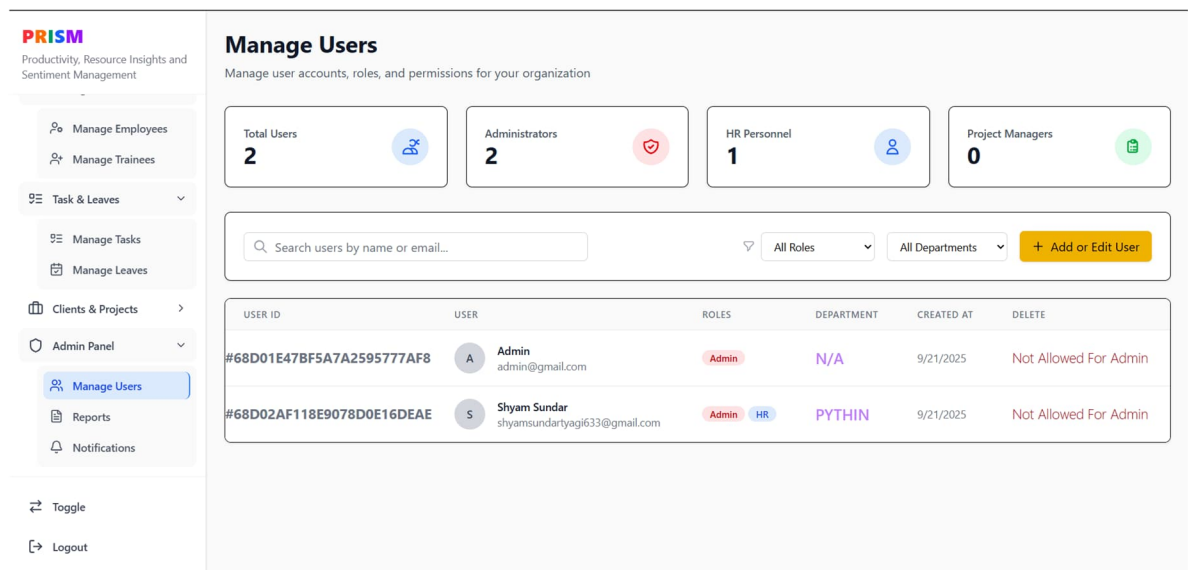


Fig. 5.3 Admin Pannel

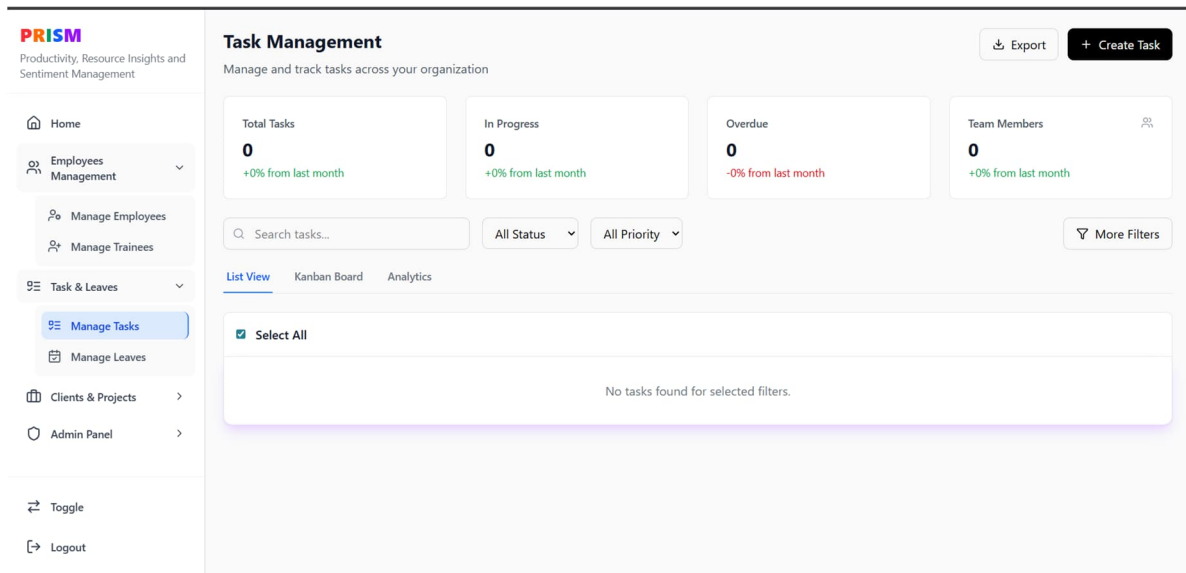


Fig. 5.4 Task Management

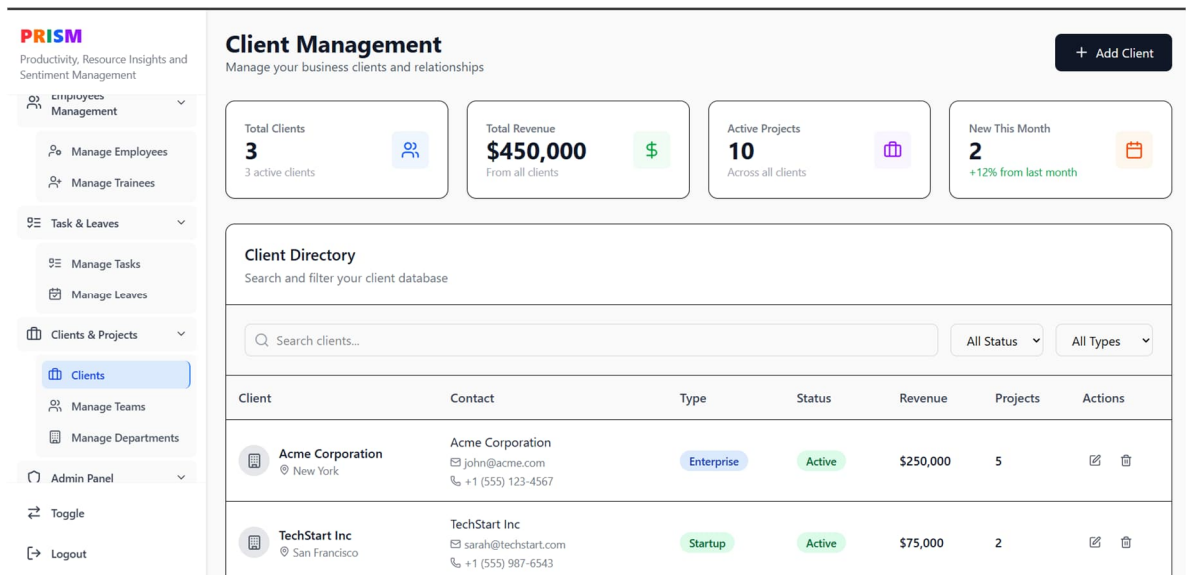


Fig. 5.5 Client Management

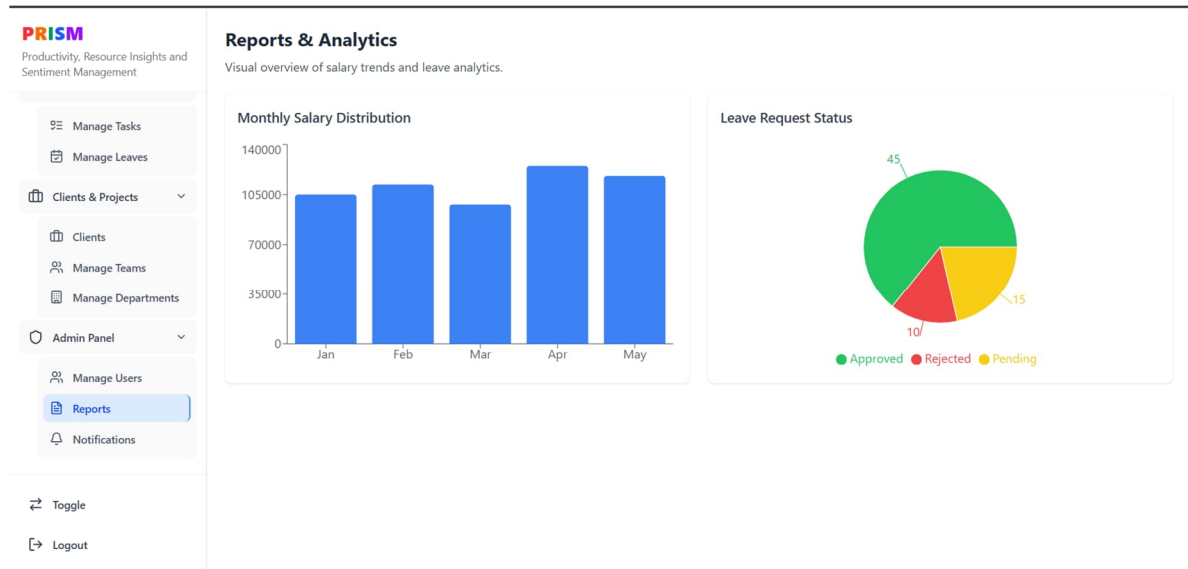


Fig. 5.6 Report And Analysis

PRISM
Productivity, Resource Insights and Sentiment Management

Manage Employees
Manage your organization's employee records

Export Add Employee

Search & Filter
Find employees using search and filters

Search by name, email, or ID... All All Status

Employees (1)
A list of all employees in your organization

EMPLOYEE ID	NAME	EMAIL	DEPARTMENT	POSITION	STATUS	HIRE DATE	ACTIONS
68D02AF18E9078D0E16DEAE	Shyam Sundar	shyamsundartyagi633@gmail.com	PYTHIN	Software developer	Active	8/12/2025	View Edit Bank De

Navigation: Home, Employees Management (Manage Employees, Manage Trainees), Task & Leaves (Manage Tasks, Manage Leaves), Clients & Projects (Clients, Manage Teams), Toggle, Logout.

Fig 5.7 Manage Employees

5.3 Detailed Design

The detailed design phase converts the high-level architectural and interface designs of the **Office ERP System – Employee Management System** into precise technical specifications that developers use during implementation. This phase defines how each module will be built, how data will be stored, and how different system components will interact.

In this phase, database structures are designed to store employee records, attendance logs, leave details, payroll information, and performance appraisal data. Each entity such as employees, departments, attendance, and payroll has clearly defined fields and relationships to ensure data consistency and accurate reporting.

The system also specifies API endpoints that allow the frontend (built using React.js) to communicate with the backend (built using Node.js and Express.js). These APIs handle functions such as user authentication, employee data updates, leave requests, payroll calculations, and report generation. JSON Web Tokens (JWT) are used to secure these APIs and ensure that only authorized users can access protected resources.

Business logic is also defined in this phase. This includes algorithms for salary calculation based on attendance and leave, updating leave balances, generating payslips, and evaluating employee performance. These rules ensure that all HR processes are automated and consistent across the system.

Performance considerations are also included in the detailed design. Efficient database queries, optimized API responses, and proper indexing are used to ensure fast data access and smooth system performance, even when many users are active at the same time.

By providing clear and detailed technical specifications, this phase ensures that the **Office ERP System – Employee Management System** can be developed accurately, efficiently, and in line with its functional and quality requirements.

5.4 Design Conclusion

In conclusion, the design phase of the **Office ERP System – Employee Management System** includes architectural design, interface design, and detailed design, each playing an important role in building a reliable and efficient ERP solution. The architectural design defines the overall system structure, the interface design ensures a user-friendly experience, and the detailed design provides clear technical guidelines for implementation.

CHAPTER 6

ARCHITECTURE

6.1 System Architecture

The architecture of the **Office ERP System – Employee Management System** is designed to provide a secure, scalable, and efficient framework for managing employee-related operations. The system follows a **three-tier architecture** consisting of the **Presentation Layer**, **Business Logic Layer**, and **Data Access Layer**. This layered approach ensures separation of concerns, making the system easier to maintain, update, and scale.

The **Presentation Layer** represents the user interface of the ERP system. It is developed using **React.js** along with **Tailwind CSS** to provide a responsive and interactive experience. This layer allows HR administrators, managers, and employees to interact with the system by accessing dashboards, viewing employee records, submitting leave requests, checking attendance, and downloading payslips. The interface is designed to be simple, intuitive, and accessible on different devices.

The **Business Logic Layer** is responsible for processing all core operations of the ERP system. It is built using **Node.js** and **Express.js**, which handle routing, API requests, authentication, and business rules. This layer processes requests coming from the frontend, performs tasks such as validating user credentials, calculating payroll, approving leave requests, and managing employee data, and then communicates with the data layer to store or retrieve information. The use of **JWT-based authentication** ensures secure and role-based access.

The **Data Access Layer** manages the storage and retrieval of data. The system uses **MongoDB** and **MySQL** to store employee records, attendance logs, leave details, payroll

data, and performance appraisal information. This layer abstracts database operations from the business logic, ensuring secure, consistent, and efficient data handling.

This three-layer architecture ensures that the **Office ERP System – Employee Management System** is reliable, scalable, and capable of supporting future organizational growth.

Layer	Description	Technologies Used
Presentation Layer	Provides the user interface for HR, managers, and employees to interact with the system.	React.js, Tailwind CSS
Business Logic Layer	Handles core system operations, business rules, and API processing.	Node.js, Express.js, JWT
Data Access Layer	Manages data storage and retrieval for employees, attendance, payroll, and performance records.	MongoDB

Table 6.1 System Architecture

6.2 Technical Architecture

The technical architecture of the **Office ERP System – Employee Management System** defines how different technologies and services work together to support employee and HR operations. The system follows a **modular service-oriented architecture**, where different modules such as authentication, employee management, attendance, payroll, and performance management operate as independent but interconnected components.

Each module communicates through **RESTful APIs**, allowing the frontend and backend to exchange data in a secure and structured manner. This architecture ensures flexibility and makes it easier to update or improve individual components without affecting the entire system. For example, the payroll module can be modified without impacting the attendance or employee management modules.

The system uses **JSON Web Tokens (JWT)** for authentication and session management. When a user logs in, a secure token is generated and used to verify all future requests. This ensures that only authorized users can access protected resources. Data

transmitted between the client and server is encrypted using **SSL/TLS**, protecting sensitive employee and payroll information.

The backend services are developed using **Node.js and Express.js**, which provide high performance and scalability. The databases (**MongoDB and MySQL**) store structured and unstructured employee-related data, ensuring fast retrieval and reliable storage.

The ERP system can be hosted on cloud platforms, allowing it to scale based on organizational needs. Cloud hosting ensures high availability, data backup, and reliable access for users working from different locations.

Overall, the technical architecture ensures that the **Office ERP System – Employee Management System** is secure, scalable, and capable of supporting real-time employee management and HR operations efficiently.

6.3 Database Architecture

The database architecture of the **Office ERP System – Employee Management System** is designed to efficiently manage all data related to employees, attendance, leave, payroll, and performance records. The system uses a combination of **MongoDB and MySQL** to store both structured and semi-structured data in a secure and scalable manner.

The database schema consists of several key collections and tables such as **Users, Employees, Attendance, Leave, Payroll, and Performance**. Each entity is designed with specific fields to store essential information required for HR and employee management. For example, the **Users** collection stores login credentials and roles, while the **Employees** collection stores personal and job-related details.

The **Employees** collection includes fields such as employee ID, name, email, department, job title, joining date, and contact information. The **Attendance** records store daily login and logout times along with attendance status. The **Leave** collection stores leave requests, types, approval status, and remaining leave balances. The **Payroll** table maintains salary details, deductions, and payslip data. The **Performance** collection stores appraisal scores and evaluation remarks.

To ensure high performance and fast data retrieval, **indexes** are created on frequently searched fields such as employee ID, email, and date. This design allows quick access to employee records, attendance history, and payroll information while maintaining data accuracy and integrity across the ERP system.

CHAPTER 7

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